Project Check-In Report
CMPE 255

Explainable Mortgage Lending Risk Intelligence Using U.S. HMDA Data

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Dataset	2024 HMDA Data (CFPB) — Conventional home purchase loans, California
GitHub / Drive	Repository created : **https://github.com/ragwort15/255_hmda-mortgage-risk-intelligence-**

1. Finalized Project Specifications

Following the initial proposal feedback, we have narrowed our scope to focus exclusively on 2024 California conventional home-purchase loans. This provides a robust dataset of approximately **159,730 records** (85,000 after specific cohort filters) to analyze the U.S. mortgage market's largest segment.

- **Objective:** To build a transparent machine learning pipeline that predicts loan outcomes (Approved vs. Denied) and utilizes SHAP to make these decisions interpretable for regulators and applicants.
- **Target Variable:** Binary classification where action_taken = 1 (Originated) vs. action_taken = 3 (Denied).
- **Modeling Strategy:** We are benchmarking a baseline **Logistic Regression** against ensemble methods, **Random Forest** and **XGBoost**.
- **Fairness Commitment:** We have incorporated a fairness analysis across race and gender groups to ensure our model aligns with the Fair Housing Act and ECOA regulatory standards.

Updated Team Roles

Member	Finalized Role
Asha Vinodhini Vijayan	Data acquisition, preprocessing, feature engineering, EDA, and fairness analysis.
Sreya Somisetty	Baseline logistic regression, evaluation metrics, and benchmarking.
Renuka Prasad Patwari	Random Forest and XGBoost modeling, hyperparameter tuning via GridSearchCV.
Anupama Singh	SHAP explainability, visualizations, business interpretation, and report integration.

2. Abstract

Mortgage lending decisions carry profound weight for individual financial health and societal equity. This project develops an explainable risk intelligence system using the 2024 California HMDA dataset. By comparing Logistic Regression with ensemble methods like Random Forest and XGBoost, we evaluate performance using F1-score and ROC-AUC to specifically address the 78%/22% class imbalance found in the data.

To bridge the gap between predictive power and transparency, we apply **SHAP** (SHapley Additive explanations) to reveal the primary drivers of loan denials—such as debt-to-income (DTI) and loan-to-value (LTV) ratios. The final pipeline is designed to be a "responsible AI" framework suitable for real-world financial deployment, ensuring that automated decisions remain auditable and fair.

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3. Report Outline

- Section 1 — Introduction: Background on mortgage risk, problem statement, research objectives.
- Section 2 — Related Work: Prior ML approaches in lending, SHAP vs. LIME, fair lending literature.
- Section 3 — Dataset and EDA: HMDA 2024 description, filtering decisions, class imbalance, visualizations.
- Section 4 — Feature Engineering: LTV/DTI derivation, categorical encoding, feature selection.

- Section 5 — Model Development: Logistic Regression, Random Forest, XGBoost training and tuning.
- Section 6 — Evaluation and Comparison: Accuracy, F1, ROC-AUC, confusion matrices, model trade-offs.
- Section 7 — Explainability with SHAP: Global importance plots, individual loan explanations, feature interactions.
- Section 8 — Fairness Analysis: Disparate impact across race/gender, regulatory implications.
- Section 9 — Business Insights: Risk drivers, underwriting recommendations, study limitations.
- Section 10 — Conclusion: Key findings, contributions, future work.
- References and Appendix: Bibliography, additional EDA plots, hyperparameter results, SHAP examples.

5. GitHub Repository and Google Drive

A GitHub repository and Google Drive folder have been set up for the project. The instructor and TA have been invited to both.

Item	Details
GitHub Repository	Created — folders: /data, /notebooks, /models, /reports
Google Drive	Shared with bertin.cordovadiba@sjsu.edu and saiharshitha.peddi@sjsu.edu
Access	Instructor and TA invited with Commentator access
Active Notebooks	01_HDMA_EDA.ipynb
README	Project overview, team info, setup instructions, folder guide

6. References

1. Consumer Financial Protection Bureau (CFPB). (2024). HMDA Data. https://ffiec.cfpb.gov/data-browser/
2. Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.
3. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning (2nd ed.). Springer.
4. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. ACM SIGKDD.
5. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.